



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 11

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
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Title of Assignment:	Docker Compose for Multi-Container App
Theory:	Docker Compose is a tool used to define and manage multi-container Docker applications using a single docker-compose.yml file. It allows multiple services, such as a web application and a database, to be configured, started, and managed together with a single command. Docker Compose automatically creates a network for communication between services, manages named volumes for persistent storage, and simplifies deployment by eliminating the need to start each container individually. This makes it easier to develop, test, and maintain applications that consist of multiple interconnected containers. Commands Used • Created a new project directory and navigated into it: mkdir my-multi-app cd my-multi-app • Started the multi-container application in detached mode: docker compose up -d • Verified the running containers: docker compose ps (or <i>docker ps</i> if <i>docker compose ps</i> was not available) • Stopped and removed the containers and network created by Docker Compose: docker compose down

Step 1: Create the Project Directory

```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

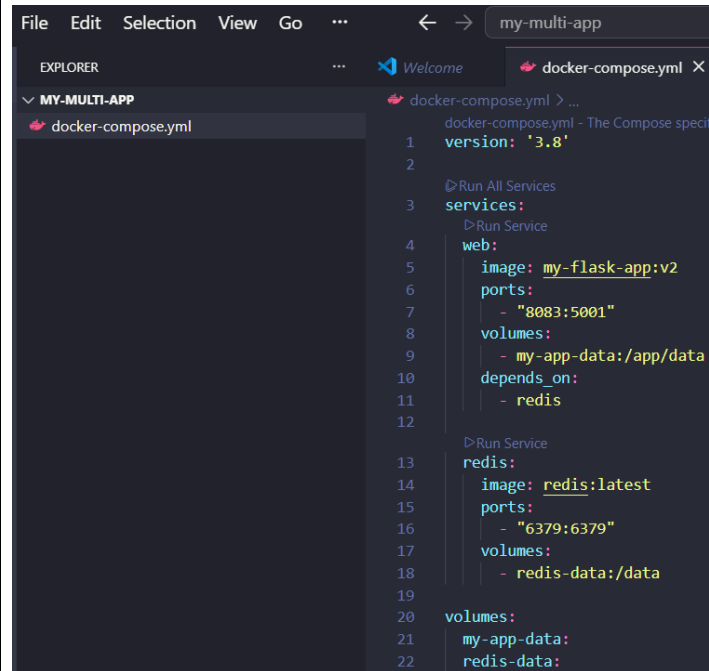
C:\Users\riavi>mkdir my-multi-app

C:\Users\riavi>cd my-multi-app

C:\Users\riavi\my-multi-app>
```

Created Project Directory — Created a new project directory named my-multi-app and navigated into it to store the Docker Compose configuration.

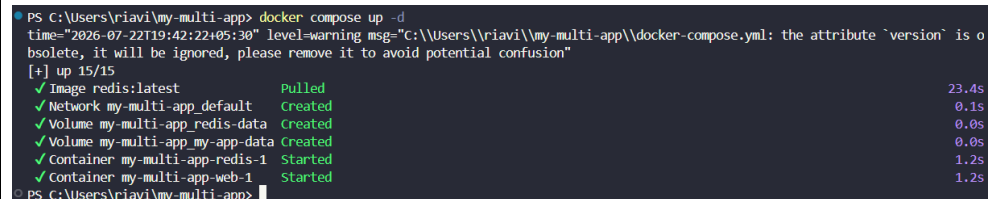
Step 2: Create docker-compose.yml



```
File Edit Selection View Go ... my-multi-app
EXPLORER
MY-MULTI-APP
  docker-compose.yml
docker-compose.yml
1 version: '3.8'
2
3 services:
4   web:
5     image: my-flask-app:v2
6     ports:
7       - "8083:5001"
8     volumes:
9       - my-app-data:/app/data
10    depends_on:
11      - redis
12
13   redis:
14     image: redis:latest
15     ports:
16       - "6379:6379"
17     volumes:
18       - redis-data:/data
19
20 volumes:
21   my-app-data:
22   redis-data:
```

Created Docker Compose Configuration — Created and saved the docker-compose.yml file defining the Flask web service, Redis service, Docker volumes, and networking configuration for the multi-container application.

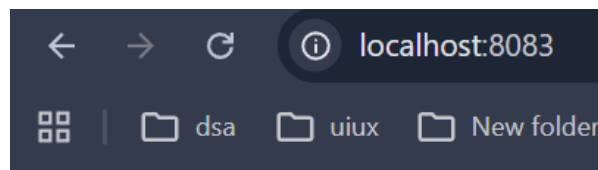
Step 3: Start the Multi-Container Application



```
PS C:\Users\riavi\my-multi-app> docker compose up -d
time="2026-07-22T19:42:22+05:30" level=warning msg="C:\\Users\\riavi\\my-multi-app\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 15/15
✔ Image redis:latest Pulled 23.4s
✔ Network my-multi-app_default Created 0.1s
✔ Volume my-multi-app_redis-data Created 0.0s
✔ Volume my-multi-app_my-app-data Created 0.0s
✔ Container my-multi-app-redis-1 Started 1.2s
✔ Container my-multi-app-web-1 Started 1.2s
PS C:\Users\riavi\my-multi-app>
```

Started Multi-Container Application — Started the multi-container application using Docker Compose, which created the required network, volumes, and launched both the Flask web service and Redis service.

Step 4: Access the Web Application



Hello, Docker!

Log:

2026-07-22 14:13:53: Hello, Docker!

Verified Web Application — Accessed the Flask application through <http://localhost:8083> and verified that the web service was successfully running as part of the Docker Compose application.

Step 5: Verify the Running Services

```
PS C:\Users\riavi\my-multi-app> docker compose ps
time="2026-07-22T19:45:11+05:30" level=warning msg="C:\Users\riavi\my-multi-app\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME                IMAGE              COMMAND                  SERVICE   CREATED   STATUS    PORTS
my-multi-app-redis-1 redis:latest       "docker-entrypoint.s..." redis      2 minutes ago Up 2 minutes 0.0.0.0:6379->6379/tcp, [
::]:6379->6379/tcp
my-multi-app-web-1   my-flask-app:v2    "python app.py"         web       2 minutes ago Up 2 minutes 0.0.0.0:8083->5001/tcp, [
::]:8083->5001/tcp
PS C:\Users\riavi\my-multi-app>
```

Verified Running Services — Verified that the Docker Compose application was running successfully by listing the active containers, including the Flask web service and the Redis service.

Step 6: Stop and Remove the Multi-Container Application

```
PS C:\Users\riavi\my-multi-app> docker compose down
time="2026-07-22T19:47:37+05:30" level=warning msg="C:\Users\riavi\my-multi-app\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] down 3/3
✓ Container my-multi-app-web-1 Removed 0.7s
✓ Container my-multi-app-redis-1 Removed 0.5s
✓ Network my-multi-app_default Removed 0.3s
PS C:\Users\riavi\my-multi-app>
```

Stopped Multi-Container Application — Stopped and removed the Docker Compose application, including the running containers and the automatically created network, using the `docker compose down` command.

Subject Faculty Name: Prof. Arijit Dutta